Phishing Message Detection using NLP

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GitHub: https://github.com/Atul013/NLP-course-project

Live application: https://phish-lens-topaz.vercel.app

Abstract

This project implements an English text classifier for educational phishing message detection. Two leakage-controlled scikit-learn pipelines compare raw TF-IDF text against meaningful NLTK preprocessing before logistic regression. On a fixed 40-message held-out test split from a small synthetic corpus, the selected NLTK pipeline obtained precision, recall, and F1 of 1.0000. That score must be interpreted cautiously: the corpus contains only 200 balanced, curated messages with strong templates and is not representative of a live inbox. The deployed Phish Lens web interface and the original Streamlit demo report the predicted class, an explicitly uncalibrated model probability, and model-derived influential terms.

1. Introduction and problem statement

Phishing messages imitate trusted entities to obtain credentials, money, or other sensitive information. Manual review does not scale, while simplistic keyword rules create false alarms. This course project asks whether a compact, interpretable NLP baseline can distinguish messages labelled phishing from messages labelled benign, while preventing common evaluation leakage.

2. Objectives

1. Use NLTK meaningfully for tokenization, lemmatization, and selective

stopword filtering while retaining negation.

2. Compare raw and NLTK-preprocessed TF-IDF logistic-regression pipelines.

3. Deduplicate and split data reproducibly before fitting text transforms.

4. Report precision, recall, F1, a confusion matrix, and error evidence.

5. Provide a reproducible notebook, tests, saved model, Streamlit demo, and deployed web interface.

3. Dataset, source, and license

The experiment uses the *Phishing and Benign Email Dataset* published on Hugging Face by Teddyha as a duplicate of darkknight25's dataset. Its dataset card describes 200 synthetic/curated English messages—100 `phishing` and 100 `benign`—and states an MIT license. Revision `ce65d43` is pinned and the raw JSONL is verified with SHA-256 `fbc5158d5308c376012a427c7a11d9a2616c6f9b569364d0e5bb085c62b0ff16`. The class labels are preserved; no spam/ham corpus is relabelled as phishing.

The subject and body are joined. Exact duplicate combined texts are removed before splitting; this run removed 0 duplicates. The final class distribution remained balanced at 100/100.

4. NLTK preprocessing

The NLTK pipeline lowercases text, replaces URLs and email addresses with explicit placeholders, uses NLTK's `TreebankWordTokenizer`, removes selected English stopwords while retaining `no`, `not`, `never`, and related negations, then applies WordNet lemmatization. Preserving negation avoids turning phrases such as “do not share” into a misleading positive phrase.

5. Methodology and architecture

The fixed random seed is 42. After deduplication, a stratified 60/20/20 split creates 120 training, 40 validation, and 40 test examples. The split identifiers are stored in `artifacts/split_ids.json`. Each scikit-learn `Pipeline` contains its own TF-IDF vectorizer and logistic-regression classifier, so the vectorizer is fitted only on training text. Features include unigrams and bigrams.

Two candidates are trained: (A) raw text with TF-IDF's lowercase handling, and (B) NLTK-preprocessed text. Highest validation F1 selects the final model; validation recall breaks a tie. The test set is not used for selection.

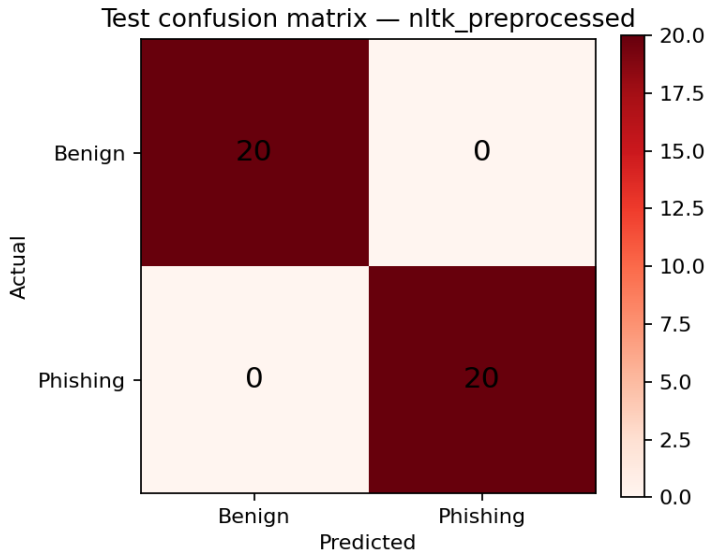
6. Implementation

`src/phishdetect/data.py` validates labels, deduplicates, and splits the data. `preprocessing.py` implements the NLTK transformation. `modeling.py` owns model construction and metrics. `inference.py` calculates predictions and term-level linear contributions. `train.py` persists auditable artifacts. `app.py` is the Streamlit interface. The notebook walks through the same public functions.

7. Experiments and actual results

Pipeline	Split	Accuracy	Precision	Recall	F1
Raw text	Validation	0.9750	1.0000	0.9500	0.9744
Raw text	Test	0.9750	0.9524	1.0000	0.9756
NLTK preprocessed	Validation	1.0000	1.0000	1.0000	1.0000
NLTK preprocessed	Test	1.0000	1.0000	1.0000	1.0000

The NLTK pipeline was selected. Its held-out confusion matrix was `[[20, 0], [0, 20]]` with zero recorded test errors. The raw baseline misclassified one benign test message as phishing and no phishing messages as benign.



8. Demo and explanation

The Streamlit demo accepts pasted text and shows `phishing` or `benign`, the model's phishing probability, and the present TF-IDF terms with the largest absolute coefficient contribution. The interface calls this an *uncalibrated model probability*, not a security risk score. These contributions describe a linear model's calculation; they do not prove why a sender wrote the message.

Phish Lens is deployed on Vercel at https://phish-lens-topaz.vercel.app. It adds a responsive interface, sample messages, processed-token inspection, keyboard controls, request validation, error handling, and cancellation of stale analyses. The server does not persist message text or fetch embedded links.

To fit the serverless bundle limit, the fitted vocabulary, IDF values, coefficients, and intercept are exported without retraining. A compact Python runtime calculates the same sublinear TF-IDF, L2 normalization, and logistic probability while retaining the original NLTK preprocessing. Verification over 20 varied inputs matches the original pipeline's probabilities and term contributions to 12 decimal places. WordNet and stopwords archives are SHA-256 verified. Thirteen API/runtime tests and live desktop/mobile browser checks passed on 14 October 2026. These are software verification checks, not additional held-out model evaluation.

9. Error analysis and limitations

The selected model produced no errors on the 40-message test set, so individual false-positive/false-negative analysis is not statistically meaningful. The `artifacts/test_errors.csv` records errors for the selected model and is empty. The raw baseline's single false positive is reflected in its confusion matrix; both pipelines' confusion matrices remain in `metrics.json`.

The major limitation is external validity. The dataset is tiny, balanced, synthetic/curated, and contains repeated stylistic templates. Random splitting can place related templates across splits, producing an optimistic score. There is no sender-header, domain-reputation, attachment, image, multilingual, temporal, or adversarial evidence. Real security notifications may resemble phishing, and new attacks can use vocabulary absent from training. The model must not autonomously block mail or guide link/attachment opening. Future work should add a larger reviewed corpus, group-aware and cross-source evaluation, temporal testing, calibrated probabilities, and human security review.

10. Conclusion

The project meets its educational objective: it demonstrates an auditable NLTK preprocessing pipeline, a leakage-controlled comparison, reproducible metrics, and an interpretable local demo. The result shows that the selected pipeline separates this specific synthetic dataset, not that phishing detection is solved. Honest scope and validation are therefore part of the deliverable.

References

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